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3.	16-2-26	Learn the basics of functions in R	
4.	23-2-26	Implement data frames in R	
5.	9-3-26	Implement different string manipulations in R.	
6.	23-3-26	Implement different Data Structures in R.	
7.	4-04-26	Write a program to read a csv file and analyze the data in the file R.	
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9.	18-04-26	Create a dataset and do statistical analysis on data using R.	

EXERCISE:01

1. **OUTPUT:**

```
> roll_no
[1] 315
> name
[1] "Divya"
> cgpa
[1] 9.12
> pass
[1] FALSE
>
```

2. **OUTPUT:**

```
> emp_id
[1] 7420
> salary
[1] 38999.25
> department
[1] "Marketing"
> is_permanent
[1] TRUE
>
```

3. **OUTPUT:**

```
> product_name
[1] "Headphones"
> price
[1] 2499
> stock_available
[1] 60
> discount_eligible
[1] FALSE
```

4. **OUTPUT:**

```
> city
[1] "Hyderabad"
> temperature
[1] 28.7
> rain
[1] TRUE
> wind_speed
[1] 6.3
>
```

5. **OUTPUT:**

```
> subject
[1] "Chemistry"
> total_marks
[1] 150
> marks_obtained
[1] 112
> result
[1] TRUE
```

6. **OUTPUT:**

```
> account_no
[1] 556677889
> holder_name
[1] "Arjun Mehta"
> balance
[1] 10500
> account_active
[1] FALSE
```

7. **OUTPUT:**

```
> movie_title
[1] "Avatar"
> rating
[1] 7.9
> reviews
[1] 875
> available
[1] TRUE
```

8. **OUTPUT:**

```
> vehicle_count
[1] 210
> location
[1] "Ring Road"
> avg_speed
[1] 55.2
> congested
[1] FALSE
```

EXERCISE:02

1. **OUTPUT:**

```
[1] "Not Eligible for Loan"
```

2. **OUTPUT:**

```
[1] "Not Eligible for Exam"
```

3. **OUTPUT:**

```
[1] "Discount Applied"
```

4. **OUTPUT:**

```
[1] "Low Usage"
```

5. **OUTPUT:**

```
[1] "Not Eligible for Bonus"
```

6. **OUTPUT:**

```
[1] "Invalid Password"
```

7. **OUTPUT:**

```
[1] "Extreme Temperature Warning"
```

8. **OUTPUT:**

```
[1] "Fail"
```

9. **OUTPUT:**

```
[1] "Credit Card Rejected"
```

10. **OUTPUT:**

```
[1] "Medium Risk"
```

11. **OUTPUT:**

```
[1] "Short Green Signal"
```

12. **OUTPUT:**

```
[1] "Not Eligible for Appraisal"
```

13. **OUTPUT:**

```
[1] "Exam Disqualified"
```

14. **OUTPUT:**

```
[1] "Normal Pricing"
```

15. **OUTPUT:**

```
[1] "Machine Shutdown"
```

16. **OUTPUT:**

```
[1] "Admission Rejected"
```

EXERCISE:03

1. **OUTPUT:**

```
[1] 2700
```

2. **OUTPUT:**

```
[1] "Fail"
```

3. **OUTPUT:**

```
[1] 40950
```

4. **OUTPUT:**

```
[1] 425
```

5. **OUTPUT:**

```
[1] 1800
```

6. **OUTPUT:**

```
[1] 71.6
```

7. **OUTPUT:**

```
[1] "Fail"
```

8. **OUTPUT:**

```
[1] "Eligible to Vote"
```

9. **OUTPUT:**

```
[1] 15279.18
```

10. **OUTPUT:**

```
[1] 8715
```

11. **OUTPUT:**

```
[1] 4680
```

12. **OUTPUT:**

```
[1] "Second Class"
```

13. **OUTPUT:**

```
[1] 900
```

14. **OUTPUT:**

```
[1] 7200
```

15. **OUTPUT:**

```
[1] 32604
```

16. **OUTPUT:**

```
[1] "High Usage"
```

EXERCISE:04

1. **OUTPUT:**

```
  RollNo   Name  Dept Marks
1    201   Asha  MECH    67
2    202 Vikram CIVIL    74
3    203 Neeraj   EEE    81
4    204  Pooja  MECH    79
```

2. **OUTPUT:**

```
  EmpID   Name    Dept Salary
1    11  Kunal   Admin  28000
2    12 Ritika Support  32000
3    13    Dev      IT  41000
4    14 Farhan      HR  36000
```

3. **OUTPUT:**

```
  PatientID   Name Age  Cost
1       301  Mohan  60 22000
2       302  Geeta  29  9000
3       303 Ramesh  47 16000
```

4. **OUTPUT:**

```
  ProductID ProductName Quantity
1        10    Notebook      200
2        20      Marker      120
3        30      Folder       75
4        40     Stapler       50
```

5. **OUTPUT:**

```
  RollNo  Name Dept Sem1 Sem2 Attendance
1     11 Irfan  BIO   65   70         85
2     12 Kavya CHEM   72   75         80
3     13 Rohit  PHY   88   90         92
4     14 Leena  BIO   78   82         88
5     15 Manoj  PHY   83   86         91
```

6. **OUTPUT:**

	EmpID	Name	Dept	Basic	HRA	DA	Rating
1	201	Sanjay	Sales	27000	5400	2700	3.9
2	202	Priya	IT	38000	7600	3800	4.6
3	203	Varun	Marketing	35000	7000	3500	4.3

7. **OUTPUT:**

	PatientID	Name	Age	Disease	Amount
1	401	Nisha	33	Asthma	8000
2	402	Ajay	58	Arthritis	14000
3	403	Kiran	46	Migraine	6000

8. **OUTPUT:**

	ProductID	ProductName	UnitsSold	Revenue
1	5	TV	45	900000
2	6	Refrigerator	30	750000
3	7	AC	25	650000
4	8	Washing Machine	20	400000

9. **OUTPUT:**

	FieldID	Crop	Moisture	Temp	ExpectedYield
1	10	Cotton	50	34	3.5
2	20	Sugarcane	42	30	5.2
3	30	Soybean	48	33	4.0

10. **OUTPUT:**

	RollNo	Name	Dept	Sem1	Sem2	Attendance
1	21	Tejas	Maths	55	60	78
2	22	Nikita	Stats	68	70	82
3	23	Harish	CS	77	80	88

11. **OUTPUT:**

	EmpID	Name	Dept	Basic	HRA	DA	Rating
1	301	Anil	IT	25000	5000	2500	4.1
2	302	Bhavna	HR	30000	6000	3000	NA
3	303	Chetan	Sales	NA	NA	NA	4.7

12. **OUTPUT:**

	PatientID	Name	Age	Disease	Amount
1	501	Deepak	40	Flu	5000
2	502	Shalini	35	<NA>	7000
3	503	Omkar	62	Cancer	20000

13. **OUTPUT:**

	ProductID	ProductName	UnitsSold	Revenue
1	11	Fan	60	300000
2	12	Cooler	40	NA
3	13	Heater	30	200000
4	14	Geyser	NA	150000

14. **OUTPUT:**

```
  FieldID   Crop Moisture Temp ExpectedYield
1     101 Barley       35   28           2.8
2     102 Millet       42   31            NA
3     103 Pulses       NA   NA           3.6
```

EXERCISE:05

1. **OUTPUT:**

```
> upper_names
[1] "KIRAN"  "SOMA"   "RAJESH"
> name_length
[1] 5 4 6
```

2. **OUTPUT:**

```
[1] "deepa" "varun" "latha"
```

3. **OUTPUT:**

```
[1] "Machine Learning" "Cloud Computing"  "Big Data"
```

4. **OUTPUT:**

```
[1]  TRUE FALSE  TRUE
```

5. **OUTPUT:**

```
[1] "issue found in module, critical issue detected"
```

6. **OUTPUT:**

```
[1] "Weak Password"
```

7. **output:**

```
[1] 6
```

8. **output:**

```
[1] "txt"  "pptx" "R"
```

9. **output:**

```
[1] "Pra" "Lak" "Div"
```

10. **output:**

```
[1] "Kiran" "Mohan" "Vijay"
```

EXERCISE:06

1. **OUTPUT:**

```
> total
[1] 395
> average
[1] 79
> highest
[1] 91
```

2. **OUTPUT:**

```
> min_temp
[1] 28
> max_temp
[1] 33
```

3. **OUTPUT:**

```
> employee$Name
[1] "Rohini"
> employee$Salary
[1] 52000
```

4. **OUTPUT:**

```
> patient$TestResults
[1] 135 128 142
```

5. **OUTPUT:**

```
  RollNo     Name  Dept Marks
1     11 Pavithra   EEE    74
2     12   Dinesh  MECH    81
3     13  Sanjana CIVIL    69
```

6. **OUTPUT:**

```
> total_sales
[1] 215000
```

7. **OUTPUT:**

```
> orders
     Item Price
1  Tablet  1500
2 Speaker  2300
3   Mouse   999
```

8. **OUTPUT:**

```
> class_results
  RollNo Total
1    201   210
2    202   225
3    203   240
```

9. **OUTPUT:**

```
> transactions
   Date Amount
1 05-02   -700
2 06-02   1500
3 07-02   -300
```

10. **OUTPUT:**

```
> records
    Subject Marks
1   English    72
2   Biology    85
3 Chemistry    78
```

EXERCISE:07

OUTPUT:

```
> Product   TotalSales
  Tablet    270000
  Camera     90000
  Shoes      22500
  Watch      16000
```

```
> Product  TotalSales
  Tablet   270000
```

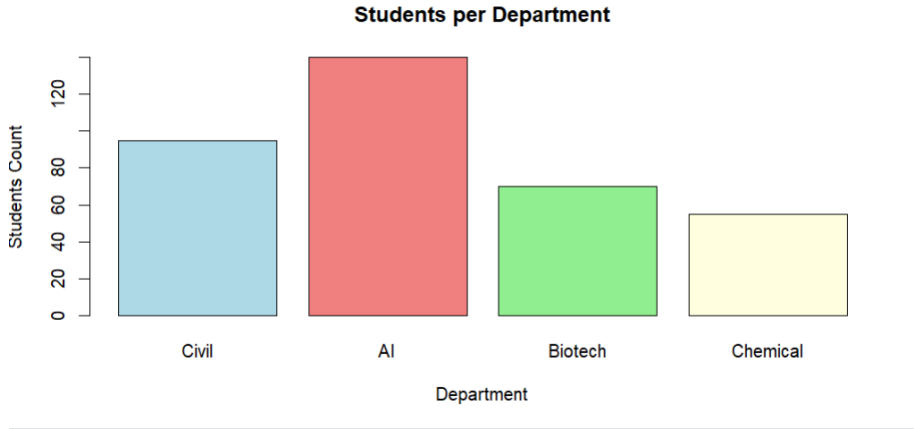
```
> high_sales <- total_sales_product[total_sales_product$TotalSales

  high_sales
```

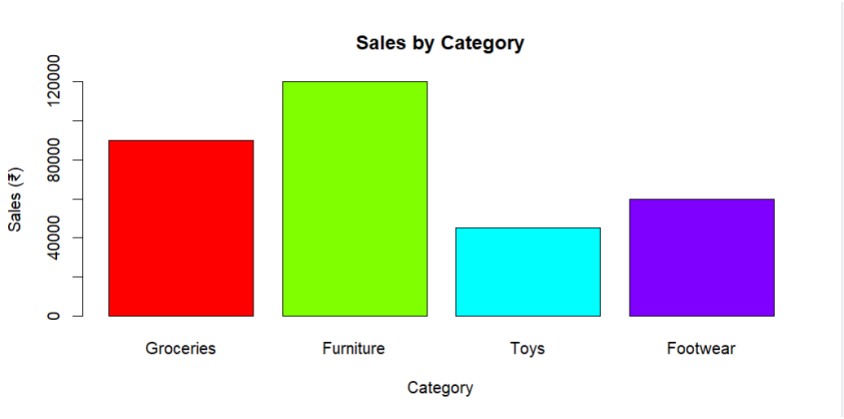
```
> Product   TotalSales
  Tablet    270000
  Camera     90000
```

EXERCISE:08

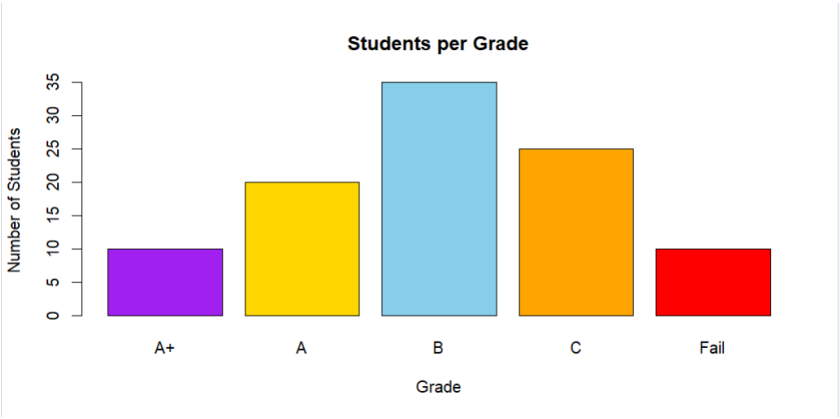
1. **OUTPUT:**



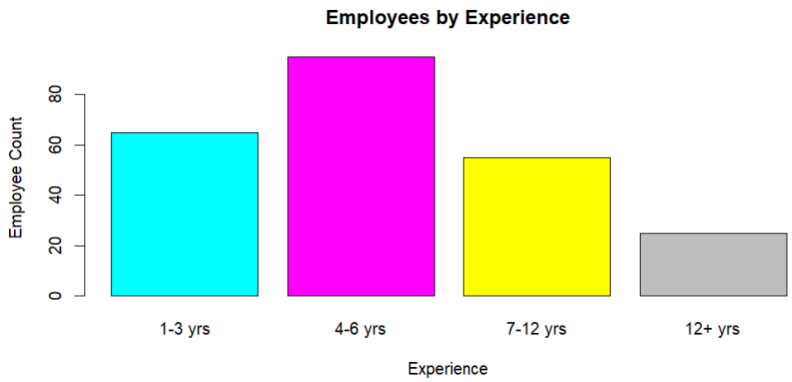
2. **OUTPUT:**



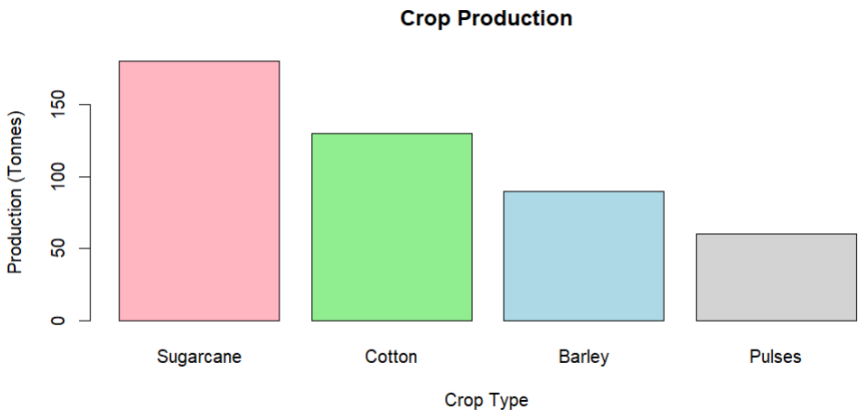
3. **OUTPUT:**



4. **OUTPUT:**



5. **OUTPUT:**



EXERCISE:09

1. **OUTPUT:**

```
> mean_marks
  Maths Science English
   79.8    78.7    80.3
> median_marks
  Maths Science English
   80.0    77.5    79.0
> sd_marks
   Maths  Science  English
9.886017 8.219895 7.703535
> top_student
   Name Maths Science English Total
8 Divya    95      93      90   278
> above_avg_math
      Name Maths Science English Total
2    Priya    84      80      78   242
4  Keerthi    91      89      94   274
6    Pooja    88      85      86   259
8    Divya    95      93      90   278
10  Swathi    82      79      81   242
```

2. **OUTPUT:**

```
> sales_stats
  Product Mean Max Min Variance
1  Tablet  4.2   6   3      1.7
2      TV  2.8   4   2      0.7
3     Fan  4.8   6   4      0.7
4   Shoes  7.6   9   6      1.3
5 Printer  1.4   2   1      0.3
6  Router  5.8   7   5      0.7
7 Monitor  5.4   7   4      1.3
> max_avg_product
  Product Mean Max Min Variance
4   Shoes  7.6   9   6      1.3
```

3. **OUTPUT:**

```
> mean_age
[1] 47.8
> median_age
[1] 49
> range_age
[1] 29 62
> sd_age
[1] 11.40955
> older_patients
   PatientID   Name Age
2        302 Bhavya  58
4        304  Deepa  62
6        306  Farha  51
8        308   Hema  55
10       310  Jyoti  60
```

4. **OUTPUT:**

```
> dept_stats
     Dept Salary.Mean Salary.Median
1 Finance    46000.00         46000
2      HR    31500.00         31500
3      IT    40666.67         41000
4   Sales    33000.00         35000
> high_earners
   EmpID    Name    Dept Salary
2    202   Leela      IT  42000
5    205   Omkar Finance  47000
6    206 Pallavi      IT  39000
8    208   Rekha Finance  45000
10   210   Tanvi      IT  41000
```

5. **OUTPUT:**

```
> mean_rating
[1] 4.1
> median_rating
[1] 4
> mode_rating
[1] 4
> var_rating
[1] 0.5444444
> top_rated
  ProductID Product Rating
1        11      TV      5
5        15    Iron      5
9        19  Vacuum      5
```